Rosh Guadiana Páez

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About Me

Biomedical Engineering student with a robust analytical foundation in pure mathematics, computational modeling, and nonlinear dynamical systems. My expertise lies in biosignal processing, full-stack application development, and the integration of artificial intelligence in healthcare. I am highly focused on solving complex problems in neuroengineering.

Academic Background

Universidad de las Américas Puebla (UDLAP) San Andrés Cholula, Puebla
Bachelor of Science in Biomedical Engineering Expected Graduation: 2028

- **GPA:** 9.7/10 — **Funding:** Awarded a 100% Academic Scholarship for sustained academic excellence.
- **STEM Representative Team:** Active member of the official UDLAP STEM team, contributing to institutional technological projects and science communication.
- **Advanced Training:** Selected institutional representative and participant in the *Putnam Mathematical Competition* preparatory program (December 2026).
- **Relevant Coursework:** Ordinary Differential Equations, Chemical Thermodynamics, Calculus II, Electronics.

Neuroengineering & Technical Projects

Phase Space Attractor Reconstruction (EEG) | Dashboard *Computational Neuroscience*

- Analyzed Electroencephalography (EEG) time-series data using nonlinear dynamical systems theory.
- Reconstructed phase space attractors utilizing Takens' embedding theorem to map the geometric structure and chaotic dynamics of neurological activity.

Hackathon por Amor a Puebla *Full-Stack & Accessibility Developer* 2026

- Developed *Pipo* (github.com/RoshTzsche/Pipo), an automated medical classification application, handling the complete end-to-end technical implementation.
- Built backend architecture with extensive API integrations, programmed the interactive frontend interface utilizing modern web frameworks, and ensured strict accessibility compliance for healthcare users.

Embedded Signal Processing & PPG Sensor Design *Hardware Architecture*

- Designed and implemented analog circuitry for a Photoplethysmography (PPG) sensor to acquire cardiovascular pulse data.
- Developed codebases for PIC16F1717 microcontrollers to execute digital filtering and signal conditioning, optimizing the signal-to-noise ratio of the acquired physiological data.

NASA Space Apps Challenge *Developer & Data Analyst* Two Consecutive Years

- **Asteroid-DB (Recent Edition):** Developed a platform (github.com/RoshTzsche/Asteroid-DB) integrating and analyzing data from Near-Earth Object (NEO) programs to model astrophysical trajectories.
- **ExoSky (Previous Edition):** Modeled and analyzed exoplanet data and celestial mechanics.

Research Experience

Undergraduate Researcher in Metagenomics *Universidad de las Américas Puebla*
Project: Dietary Metagenomics of the Narrow-Clawed Crayfish 2026 – Present

- Implement optimized bioinformatics pipelines in Python and R for taxonomic classification and multidimensional diversity modeling (PCoA) under the supervision of Dr. Osiris Díaz Torres.

Teaching, Outreach & Science Communication

STEMpire Podcast (UDLAP STEM Team) *Interviewer, Editor & Guest Speaker*
Science Communicator Multiple Appearances

- Core contributor to the official podcast of the UDLAP STEM Representative Team, editing episodes and engaging in scientific dissemination regarding biomedical engineering and research methodologies.

Verano STEM *Universidad de las Américas Puebla (UDLAP)*
Guest Lecturer & Instructor 2025 – Present

- **Upcoming (Summer 2026):** Scheduled to deliver specialized lectures on Quantum Computation and Pure Mathematics for the institutional STEM outreach program.
- **Previous Editions:** Conducted instructional sessions and lectures focusing on the structural biology and fundamentals of DNA.

Noche de las Estrellas *Workshop Instructor* 2024 & 2025

- Designed and facilitated interactive science workshops for massive public events aimed at astronomical and scientific dissemination.

Conferences, Olympiads & Additional Experience

- **Mathematical Competitions:** Active participant in the *Olimpiada Mexicana Universitaria de Matemáticas* (OMUM 2026) and upcoming participant in the *Pierre Fermat* competition (2026).
- **MUTVI 4:** Presented original computational work on the reconstruction of phase space attractors from Electroencephalography (EEG) signals.
- **Swift Social Change Makers Enactus (2025):** 2nd Place National Award for designing high-impact AI technological solutions.
- **National Physics Olympiad (2023):** Selected for the Durango State Delegation to compete in the National phase.
- **XXXII National Chemistry Olympiad (2022):** Awarded the National Bronze Medal.
- **State Biology Olympiad (2020):** Awarded the State Gold Medal.
- **COCYTED Advanced Training (2017 – 2021):** Selected by the State Council of Science and Technology of Durango (COCYTED) for intensive extracurricular training in biology, chemistry, and medicine.

Skills and Competencies

- **Programming & Web Stack:** Python, C, R, Swift, React 18, Node.js, Bash.
- **Advanced Mathematics (Independent Study):** Real analysis, general topology, measure theory ($\mathscr{L}^p$ spaces), formal logic, and nonlinear dynamical systems theory.
- **Engineering & Bioinformatics:** EEG data analysis, analog circuit design, digital signal processing, Kraken pipelines.
- **Working Environment:** Arch Linux ecosystem (CachyOS, Hyprland), Neovim, Git, LaTeX / Typst.
- **Languages:** Spanish (Native), English (Advanced), French (Basic).